TDA Analysis of GLMP Biological Processes — One-Page Summary

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What We Did

We applied topological data analysis (TDA) to 108 genetic regulatory circuits from the Genome Logic Modeling Project (GLMP). Each circuit is represented as a Mermaid flowchart with nodes, conditionals, and logic gates (OR, AND, NOT). We extracted 5 features per process, standardized them, and computed persistent homology (Vietoris-Rips, Ripser, maxdim=2).

Main Results

Homology	Count	Interpretation
H0 (components)	108	One per process
H1 (loops)	**33**	Potential feedback/cyclic structure
H2 (voids)	**4**	Higher-dimensional cavities

The 33 H1 features and 4 H2 features indicate nontrivial topological structure in the process space.

Why This Might Be Interesting

- **Domain:** Genetic circuits have inherent topology (feedback loops, regulatory cascades). TDA may reveal structure that correlates with known biology.
- **H1 peaks:** The 5 most persistent H1 features have persistence 0.23-0.33 (birth ~1.1-1.5, death ~1.4-1.8). These could correspond to circuits with distinct feedback architectures.
- **Collaboration angle:** I bring domain knowledge (which circuits have feedback vs feedforward); you bring TDA methods. The diagrams are a shared object for interpretation.

Attachments

1. **glmp_persistence_diagram.png** - Persistence diagram (H0, H1, H2)
2. **glmp_tda_report.html** — Full report with feature statistics
3. **glmp_features.csv** — Raw feature data (108 rows × 10 columns)

4. **ADDENDUM_FOR_VEJDEMO_JOHANSSON.md** — Methods, key persistence values, discussion questions

Questions I'd Like to Explore

1. Do any persistent H1 loops correspond to known feedback circuits?
2. Would graph-based or alternative features improve interpretability?
3. Would Mapper, persistent cohomology, or subgroup comparisons be natural next steps?